

Flow 2 — validation report (16-Jul)

What was tested: the new interactive Flow 2 — your flow document executed as an auditable SOP sequence with a decision point at every step, and numbers that are **pure history** (percentiles of the matched IP cases — no tariff, LOS or calculation logic anywhere). Live on the dev engine (`POST /api/flow2/evaluate`); the review UI ships as an admin "Flow 2" mode beside the existing Flow view. Nothing in the current flow was changed.

The sweep

374 test cases — every onboarded family (~170) × Cash + GIPSA, every 5th family × Non-GIPSA, plus targeted probes (gibberish wording, multi-treatment combo, mode toggles). **604 total evaluations** including answer round-trips. **215 clarifying questions were raised and answered** the way an FC would (majority option), and every answer was honored.

Outcome	Count
Cases evaluated	374
Reached numbers (fully decided)	327
— through the package path	69
— through the non-package path	258
Clarifying questions asked / answered	215 / 215
Silent wrong answers	0
Data-invariant failures	0

What was asserted, case by case

1. **Step order** always matches the flow doc: payor → family → surgical-medical / daycare / robotic characterization → billing identification (package master) → FC-historic template (the fallback ladder) → per-payor summary.
2. **Payor resolution** correct for all three payor types: Cash ⇒ TR1 (KIMS, no insurer mapping touched), GIPSA org ⇒ TR290, Non-GIPSA org ⇒ its own tariff.
3. **Questions carry the history:** every clarifying question includes the case counts ("Robotic — 61% of 248 GIPSA cases had it"), and numbers are withheld until the question is answered — the flow never guesses past an ambiguity.
4. **Numbers are really pure history:** for every case where the full IP list was returned, the gross P25/P50/P75 were **independently recomputed from the listed IP cases and matched the reported numbers exactly**. The values are the history of the named cases — nothing else.
5. **Case sets obey their filters:** every returned IP row was checked against the declared filters (payor scope, daycare, robotic, care type) — no case leaks across a filter; widening the payor scope never shrinks the set.
6. **Package-path coherence:** a package decision always carries the package code, and the historic-template step always shows which ladder rung decided it. Inclusion/exclusion text is attached as review-only — amounts come from past billing metrics, per the 12:37 call.
7. **Honest edges:** nonsense wording ⇒ "no similar treatment found in the FC historic dataset" (a stated dead end, not a forced match); "lap chole + hernia" ⇒ the combo is detected and announced.

Failures found (6 of 374) — both causes closed same day

- **5x family flip on answer round-trips** — the AI family matcher can occasionally re-rank between the question call and the answer call, flipping the cohort mid-conversation. **Fixed:** the UI now pins the resolved family on every round-

trip; a pin equal to the engine's own top match stays marked "auto" (it only reads "you chose this" when the user genuinely overrode).

- **1x transient network blip** on one call — retried fine.
- Also fixed from review: the payor step's explanation for Cash wrongly mentioned the insurer mapping ("TR290 = GIPSA") — the decision was always correct (Cash ⇒ TR1), the sentence now says so plainly.

Deviations from the flow doc (deliberate, flagged for your call)

1. **Robotic wording auto-decides:** "robotic TKR" in the treatment text sets robotic = yes without asking (same rule as the gate). Overridable.
2. **Multi-treatment combos:** detected and announced, but only the first treatment is priced in this phase — the doc's "a path for each" is the next phase of the combo work.
3. **One question at a time** rather than all ambiguities upfront — each answer can change what the next question should be.

Added same day (the three items originally deferred)

1. **mode = logic / both — the dissection layer.** The real engine build runs alongside the pure history (same decided family/package/robotic; room type selectable, Typical mode) and every bucket gets a verdict: *"Room — historically ₹36,871–₹49,208; logic produced ₹42,558 — within"*, plus a gross-total verdict. The logic build receives NO free text — it prices the audited decisions, so it can never disagree with the trail about WHAT is being priced, only about the amounts. Verified live: historic numbers are byte-identical across modes; the contracted robotic ₹1.2L rides the logic side; switching to a General room moves only the logic figures.
2. **Combo — a path per treatment.** "lap cholecystectomy + inguinal hernia repair" now evaluates each treatment through the full SOP independently — its own trail, its own questions, its own numbers — with tabs per treatment, the identified billing shape ("2 packages" / "package + non-package"), and a combined historic-P50 strip (marked as an upper-bound reference: combo interactions like shared LOS/OT are not modeled yet). Answering a question on one path changes only that path (verified).
3. **Matcher stability + speed.** The AI matching (family AND package ranking) is now cached per wording within a session with retry/backoff underneath: answer round-trips dropped from ~7s to ~0.04s, the same wording can never resolve differently mid-conversation, and the transient flake rate seen under concurrent testing retries silently instead of surfacing. Verified: three consecutive combo evaluations give the identical billing shape.

Net: the flow executes exactly as documented, every step is auditable with its evidence and case counts, the numbers are provably the history of the named IP cases — and the logic layer is now available side-by-side for the dissection, per treatment, at interactive speed. Ready for hands-on review on the test server.

Appendix — raw suite output

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Cases: 374 (evaluations incl. answers: 604) in 1227s
Questions asked: 215, answered: 215
Reached numbers: 327 · package path: 69 · non-package: 258 · no-match: 0 · matcher-flake: 46

Failures: 6
I4 (5) — answered axis not marked decided_by user (family re-rank between
calls; fixed via the family pin):
immunotherapy × Cash · uti_urosepsis_medical_management × GIPSA ·
minor_procedure_biopsy × Cash · combined_upper_gi_endoscopy_and_colonoscopy
× Cash · clavicle_fracture_fixation × Cash
HTTP (1) — turbt_transurethral_resection_of_bladder_tumor × Cash: transient
fetch failure on one answer round
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